

Harmonic Patterns SwingCore

Installation and User Manual

Part 1 — Installation

This indicator runs on the **MetaTrader 5** platform. Installation takes a few minutes and requires no technical knowledge.

Prerequisites

- MetaTrader 5 installed and running.
- The standard **Heiken Ashi** indicator available in MetaTrader's examples folder (included by default in `Indicators\Examples\Heiken_Ashi`). The indicator uses it internally as a calculation base.

Option A — Install from the MetaQuotes Market (recommended)

1. Open MetaTrader 5.
2. Open the **Navigator** window with **Ctrl+N** (or via **View** → **Navigator**). Indicators and Expert Advisors are listed there.
3. In the **Market** section, search for “**Harmonic Patterns SwingCore**”.
4. Press **Buy** (Pro version) or **Download** (free Lite version).
5. The indicator installs automatically and appears in the Navigator under **Indicators** → **Market**.

Option B — Manual installation (.ex5 file)

1. Open MetaTrader 5.
2. Go to **File** → **Open Data Folder**.
3. Navigate to: `MQL5` → `Indicators`.
4. Copy the indicator file (`.ex5`) into that folder.
5. Back in MetaTrader 5, right-click **Indicators** in the Navigator and select **Refresh**.
6. The indicator appears in the list, ready to use.

Applying the indicator to a chart

1. Open the chart of the instrument you want to analyze (for example, XAUUSD or BTCUSD).
2. Select the desired timeframe (for example, M1).
3. Open the **Navigator** with **Ctrl+N** and locate **Harmonic Patterns SwingCore** under Indicators.
4. Drag it onto the chart, or double-click it.
5. The parameter window opens. Adjust the settings following Part 2 of this manual and press **OK**.

Verifying a correct installation

- The indicator begins scanning the chart. After a few seconds, the peak-and-valley zigzag appears, along with any detected harmonic patterns.
- When loading over extensive history, the process may take a moment. If the chart has little history loaded, scroll left so MetaTrader loads more bars.

Part 2 — The Two Detection Methods

The indicator can validate patterns in two independent ways. Both scan the same swing structure and recognize the same six patterns (Gartley, Bat, Butterfly, Crab, Cypher, Shark), but they judge validity by different criteria. The **Classic:true/Geometric:false** parameter selects which one is active.

Classic method (ratio-based)

The classic method validates a pattern by measuring the **Fibonacci proportions** between its legs. For each candidate set of five points (X, A, B, C, D), it computes the ratios between consecutive legs — such as AB/XA , BC/AB , and XD/XA — and checks whether they fall within the defined tolerance of the ideal Fibonacci values that characterize each pattern.

This is the traditional, textbook definition of a harmonic pattern: a figure is a Gartley, a Bat, or a Crab because its proportions match specific numbers. It is direct and transparent — the pattern is what the ratios say it is.

Geometric method (rule-based)

The geometric method validates a pattern by its **relative structure** rather than its exact proportions. Instead of asking “is this ratio close to 0.618?”, it asks a series of positional questions: is point B above X but below A? Does C stay below A? Does D extend beyond X, or stop short of it? Each pattern has its own set of these inequalities, and a figure qualifies only if every positional rule for that pattern holds.

This approach captures the *shape* of the pattern — its topology — independently of whether the Fibonacci ratios land exactly on their ideal values. It is more permissive about precise proportions but stricter about form: the five points must be arranged in the correct spatial order for that specific pattern.

Which one to use

Neither method is objectively “correct” — they are two valid readings of the same structure. The classic method is closer to the canonical Fibonacci definition; the geometric method is closer to how the eye recognizes a shape. Try both and see which better matches the way you read the chart. In the Pro version they can be switched freely; the Lite version offers the classic method only.

Part 3 — Input Parameters

Each parameter of the configuration window is described below, grouped by section. The heading of each item matches the label shown in the indicator’s property window.

General settings

Timeframe (*detection timeframe*)

The timeframe on which detection runs. With `PERIOD_CURRENT`, the indicator uses the active chart’s timeframe. Leave as is unless you want to force a specific timeframe. Recommended: `PERIOD_CURRENT`.

Clean Objects (*clear chart objects*)

When `true`, deletes all chart objects. This is a manual cleanup utility: enable it, let it clean, then set it back to `false`. Use it if the chart has accumulated objects from previous sessions. Recommended: `false`.

Harmonics Settings

Classic:true / Geometric:false (*detection method*)

Selects the detection method. With `true`, the classic (ratio-based) method is used; with `false`, the geometric (rule-based) method is used. Both detect the same six patterns with different criteria (see Part 2). Recommended: `true`.

Note on the Lite version: in the free version the geometric method is disabled. Setting this parameter to `false` will display no patterns. The geometric method is available only in the Pro version.

Scan Harmonics (*enable pattern detection*)

Enables or disables pattern detection entirely. With `false`, the indicator shows only the structure zigzag, without marking patterns. Recommended: `true`.

Show percentages (*per-leg percentage labels*)

Displays on the chart the percentage of each leg of the most recent pattern (XA, AB, BC, CD, XD), expressed as a proportion of its reference leg. Useful for visually verifying the ratios. Recommended: `true`.

Show horizontal line (*horizontal line from D*)

Draws a horizontal line projected from point D (pattern completion) into the future. Serves as a visual reference of the price level where the pattern completed. Recommended: `true`.

Show Fibonacci targets (*Fibonacci projection levels*)

Displays the Fibonacci projection levels of the most recent pattern, both in the pattern's direction (direct) and on the opposite side (inverse or mirror). Projected levels are: 38.2%, 50.0%, 61.8%, 100.0%, 161.8%, 261.8%, and 361.8%. Recommended: **true**.

Ratio tolerance (*ratio tolerance*)

Defines how much deviation is allowed in the ratios for a pattern to be considered valid. A lower value (for example, 0.05) makes detection stricter: fewer but more precise patterns. A higher value (for example, 0.20) is more permissive: more patterns, but less closely matched to the ideal proportion. Default: 0.15.

Alerts Settings

Enable alerts (*master alert switch*)

Master switch for the alert system. With **false**, no alert is issued even if the sub-settings are active. Recommended: **true**.

Visual pop-up (*visual pop-up alert*)

Shows a pop-up window in the terminal when a new pattern appears. Recommended: **true**.

Sound alert (*sound alert*)

Plays a sound when a new pattern appears. Requires the sound file to be available (see next parameter). Recommended: your preference.

Audio file alert (*sound file name*)

Name of the sound file played with the audio alert. The file must be in MetaTrader's **Sounds** folder. Default: **alert.wav** (included with the platform).

Alert repeats (*alert repeat count*)

Number of times the alert repeats when a new pattern appears. The alert sounds on the bar where the pattern appears and repeats on the following bars until this count is reached, then stops automatically. Default: 5.

ZigZag Settings

Windows Back (*scan window (bars back)*)

Number of bars back the indicator analyzes to build the structure and search for patterns. A larger value shows more history and older patterns, at the cost of heavier computation. A smaller value focuses on recent data and is lighter. Default: 21600.

Ratio prominence (*minimum prominence*)

Controls how pronounced a price turn must be to count as a valid peak or valley. A higher value filters out small turns (fewer points, cleaner structure); a lower value is more sensitive (more points, more noise). This is one of the key parameters for tuning detection sensitivity. Default: 2.0.

Usage Tips

- **Start with the defaults.** They are calibrated for general use. Adjust *Ratio tolerance* and *Ratio prominence* only once you understand how they affect detection.

- **Structure comes first.** If the zigzag does not reflect the turns you consider relevant, adjust *Ratio prominence* before anything else: pattern quality depends on the quality of peak-and-valley detection.
- **More tolerance is not better.** Increasing *Ratio tolerance* produces more patterns, but less faithful to the ideal proportion. Fewer well-formed patterns are usually more useful than many approximate ones.
- **Convergence matters.** When several patterns coincide in the same price area, that area gains relevance as a possible support or resistance.

Disclaimer

This indicator is a technical-analysis tool provided for informational and demonstrative purposes only. It does not constitute financial advice or any trading recommendation. Pattern detection is the result of a geometric calculation and does not represent a prediction of future market behavior. Past performance does not guarantee future results. Trading financial markets carries a high risk of loss. **You assume the entire risk of any decision you make based on this indicator.**